Heterogeneous Beliefs and Financial Markets

Based on Blume and Easley (2006)
QuantEcon Lecture 23

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Overview

Central question: *"If you're so smart, why aren't you rich?"* — Blume and Easley (2006)

Blume and Easley study how **heterogeneous beliefs** about income-process probabilities influence:

- consumption allocations
- prices of stocks, bonds, and insurance policies
- the long-run distribution of wealth among agents

Historical backdrop: Alchian (1950) and Friedman (1953) conjectured that competitive asset markets reward better-informed traders, transferring wealth to those with more realistic probability models.

This lecture provides a formal, tractable example with **complete markets**.

Roadmap

Two institutional arrangements — same outcome:

1. **Perfect socialism** — a benevolent central planner dictates history-contingent consumption allocations.
2. **Competitive markets** — selfish price-taking agents trade contingent claims at equilibrium prices.

The **two fundamental theorems of welfare economics** guarantee that these arrangements produce *identical allocations* once the planner's Pareto weights are set appropriately.

Key tool: a **likelihood ratio process** links agents' relative beliefs to their relative consumption shares across time and states.

Review: Likelihood Ratio Processes

Setup. A nonnegative random variable W is drawn i.i.d. from *either* f *or* g.

- Nature chooses the density *once and for all* before time begins.
- Observers know both f and g but *not* which one nature chose.
- We observe $\{w_t\}_{t=1}^{T}$ and want to infer nature's choice.

Period-t likelihood ratio:

$$\ell(w_t) = \frac{f(w_t)}{g(w_t)}, \quad t \geq 1.$$

Under density g, $\ell(w_t)$ is nonnegative with $\mathbb{E}_g[\ell(w_t)] = 1$.

Likelihood ratio process:

$$L(w^t) = \prod_{i=1}^{t} \ell(w_i), \quad L(w^t) = \ell(w_t)\, L(w^{t-1}).$$

Beta-Distribution Example

Concrete densities used throughout:

Density	Distribution	Parameters
f	Beta	$(\alpha, \beta) = (1,\ 1)$ (uniform)
g	Beta	$(\alpha, \beta) = (3,\ 1.2)$

- Both densities put positive probability on the same interval $(0, 1)$.
- Simulation: draw $w_t \sim g$ and compute $L_t = \prod_{i=1}^{t} \ell(w_i)$.
- By the **law of large numbers**, $\log L_t \to -D_{\mathrm{KL}}(g \| f) < 0$ almost surely when data come from g, so $L_t \to 0$.

Blume and Easley's Setting

Aggregate endowment. At each date $t = 0, 1, 2, \ldots$, a single nonstorable consumption good is available in aggregate quantity *one*:

$$y_t^1 + y_t^2 = 1.$$

- $s_t \in (0, 1)$: i.i.d. draw from Beta(θ) — *nature's density* $\pi(s_t)$.
- Agent 1's endowment: $y_t^1 = s_t$.
- Agent 2's endowment: $y_t^2 = 1 - s_t$.

History. $s^t = (s_t, s_{t-1}, \ldots, s_0)$, joint density $\pi^t(s^t) = \pi(s_t) \cdots \pi(s_0)$.

Feasibility: $c_t^1(s^t) + c_t^2(s^t) = 1$ for all s^t, all $t \geq 0$.

Nature and Agents' Beliefs

Nature draws i.i.d. from its own density $\pi(s_t)$.

Agents have heterogeneous beliefs:

- Agent i believes nature draws from $\pi^i(s_t)$.
- Agent i is *mistaken* unless $\pi^i = \pi$.
- A **rational expectations model** would impose $\pi^i = \pi$ for all i.

Our focus:

- Agent 1 believes marginal density $\pi^1(s_t) = f(s_t)$.
- Agent 2 believes marginal density $\pi^2(s_t) = g(s_t)$.
- Nature uses $\pi(s_t) = h(s_t)$, possibly a mixture of f and g.

Common discount factor: $\delta \in (0, 1)$.
Common one-period utility: $u(c) = \ln(c)$.

Intertemporal Utilities

Agent i's intertemporal expected utility:

$$V^i = \sum_{t=0}^{\infty} \sum_{s^t} \delta^t \, u\big(c_t^i(s^t)\big) \, \pi_t^i(s^t). \tag{1}$$

- $\pi_t^i(s^t) = \pi^i(s_t) \cdots \pi^i(s_0)$ is agent i's joint density over histories.
- Agent i ranks consumption plans using *their own subjective beliefs*.

Social welfare criterion (planner with Pareto weight $\lambda \in [0, 1]$):

$$\mathcal{W} = \lambda \, V^1 + (1 - \lambda) \, V^2.$$

Setting $\lambda = 0.5$ expresses *egalitarian* social preferences.

Social Planner's Allocation Problem

Maximise $\mathcal{W} = \lambda V^1 + (1-\lambda) V^2$ subject to $c_t^1(s^t) + c_t^2(s^t) = 1$.

First-order conditions (dividing agent 1's FOC by agent 2's):

$$\frac{\pi_t^2(s^t)}{\pi_t^1(s^t)} \cdot \frac{1/c_t^2(s^t)}{1/c_t^1(s^t)} = \frac{\lambda}{1-\lambda} \quad \Longrightarrow \quad \frac{c_t^1(s^t)}{c_t^2(s^t)} = \frac{\lambda}{1-\lambda}\, l_t(s^t),$$

where the **likelihood ratio**

$$l_t(s^t) = \frac{\pi_t^1(s^t)}{\pi_t^2(s^t)} = \prod_{i=0}^{t} \frac{f(s_i)}{g(s_i)}.$$

The Planner's Allocation Rule

Using feasibility $c_t^1 + c_t^2 = 1$, the planner's rule is:

$$\boxed{c_t^1(s^t) = \frac{\lambda\, l_t(s^t)}{1 - \lambda + \lambda\, l_t(s^t)}.} \tag{2}$$

Continuation Pareto weights. Define

$$\lambda_t(s^t) \;=\; \frac{\lambda\, l_t(s^t)}{1 - \lambda + \lambda\, l_t(s^t)},$$

so $c_t^1(s^t) = \lambda_t(s^t)$ and $c_t^2(s^t) = 1 - \lambda_t(s^t)$.

- The continuation Pareto weight evolves endogenously with the likelihood ratio process.
- A *smarter* agent (one whose beliefs better match nature's) sees $l_t \to \infty$ and therefore $\lambda_t \to 1$.

"If you're so smart, why aren't you rich?"

Three limiting cases for $l_\infty = \lim_{t\to\infty} l_t(s^\infty)$:

Case	Meaning	l_∞	c^1_∞
1	Both agents equally (un)smart	1	λ
2	Agent 2 is smarter	0	0
3	Agent 1 is smarter	∞	1

- When beliefs are *equally wrong*, consumption shares stay put at $(\lambda, 1-\lambda)$.
- When one agent is smarter, that agent eventually *owns the entire endowment stream*.
- This is Alchian/Friedman, made precise.

Competitive Equilibrium with Complete Markets

Setup. At time -1, a complete set of **Arrow–Debreu** markets open.

- Price $p_t(s^t)$: cost at $t = -1$ of one unit of consumption delivered at date t after history s^t.
- Each agent takes prices as given and chooses $\{c_t^i(s^t)\}$ to maximise V^i subject to:

$$\sum_{t \geq 0} \sum_{s^t} p_t(s^t)\, c_t^i(s^t) \;\leq\; \sum_{t \geq 0} \sum_{s^t} p_t(s^t)\, y_t^i(s^t).$$

FOC for agent i (attaching Lagrange multiplier μ^i):

$$p_t(s^t) = \frac{\delta^t\, \pi_t^i(s^t)}{\mu^i\, c_t^i(s^t)}.$$

Equilibrium Prices

Guess-and-verify. Matching the competitive equilibrium with the planning solution by setting $\mu^1 = 1 - \lambda$ and $\mu^2 = \lambda$:

$$\boxed{p_t(s^t) = \delta^t \, \lambda(1-\lambda)\Big[(1-\lambda)\,\pi_t^2(s^t) + \lambda\,\pi_t^1(s^t)\Big].}$$

Limiting cases:

- $l_\infty = 0$ (agent 2 smarter): tails of prices reflect *agent 2's* model, $p_t(s^t) \propto \delta^t \pi_t^2(s^t)$.
- $l_\infty = \infty$ (agent 1 smarter): tails of prices reflect *agent 1's* model, $p_t(s^t) \propto \delta^t \pi_t^1(s^t)$.
- For small t: prices reflect a *weighted mixture* of both models.

Conclusion: competitive equilibrium prices converge to reflect the beliefs of the *smarter* agent — Alchian/Friedman confirmed.

Simulation Design

Belief distributions: $f = \text{Beta}(1,1)$ (uniform), $\quad g = \text{Beta}(3, 1.2)$.

Three scenarios for nature's density h:

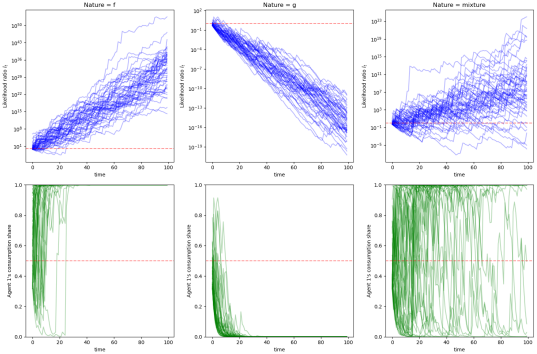
Scenario	Nature draws from ...
Left	f (same as agent 1)
Middle	g (same as agent 2)
Right	$h = 0.5f + 0.5g$ (mixture)

Parameters: $\lambda = 0.5$, $T = 100$ periods, $N = 10{,}000$ simulated paths.

Allocation rule implemented:

$$c_t^1 = \frac{\lambda L_t}{1 - \lambda + \lambda L_t}, \quad L_t = \prod_{i=1}^{t} \frac{f(s_i)}{g(s_i)}.$$

Simulation Results: Likelihood Ratios and Consumption Shares



Top row: log-scale likelihood ratio paths for each scenario. **Bottom row:** agent 1's consumption share paths ($\lambda = 0.5$ dashed).
Left: nature$= f$ — agent 1 quickly consumes everything. Middle: nature$= g$ — agent 1's share drifts to zero (more slowly). Right: nature$= h$ — mostly resembles the left panel since $D_{\mathrm{KL}}(h\|g) > D_{\mathrm{KL}}(h\|f)$.

KL Divergences: Key Numbers

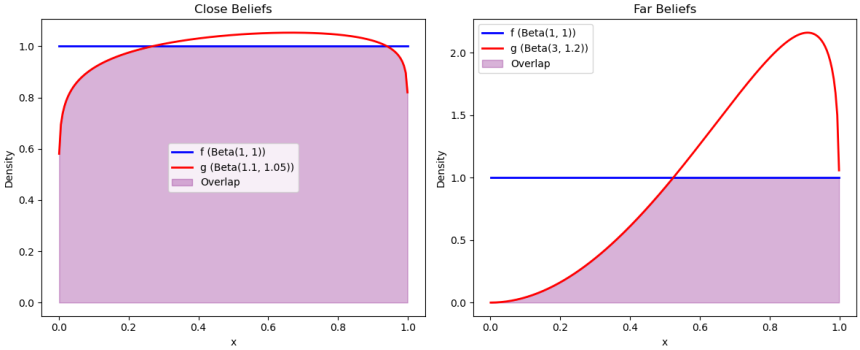
Kullback–Leibler divergence governs the rate of learning:

$$D_{\mathrm{KL}}(p\|q) = \int p(w)\ln\frac{p(w)}{q(w)}\,\mathrm{d}w \;\geq\; 0.$$

Divergence	Value	Implication
$D_{\mathrm{KL}}(f,g)$	0.759	large surprise when one holds g and nature is f
$D_{\mathrm{KL}}(g,f)$	0.344	moderate surprise in reverse case
$D_{\mathrm{KL}}(h,f)$	0.073	h is close to f
$D_{\mathrm{KL}}(h,g)$	0.281	h is further from g

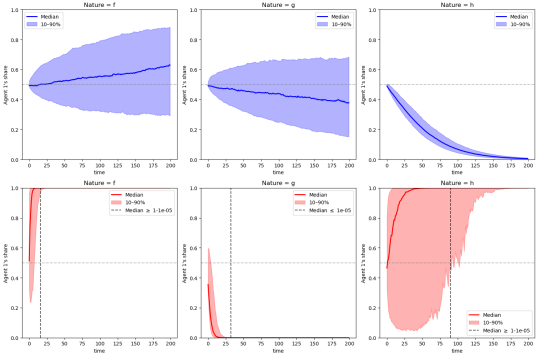
Since $D_{\mathrm{KL}}(h,g) > D_{\mathrm{KL}}(h,f)$, agent 1's beliefs ($f$) are *closer* to nature's mixture (h) than agent 2's beliefs (g).

Close vs. Far Beliefs: Distributions



Left: "Close" beliefs: $f = \text{Beta}(1,1)$, $g = \text{Beta}(1.1, 1.05)$ — large overlap (purple).
Right: "Far" beliefs: $f = \text{Beta}(1,1)$, $g = \text{Beta}(3, 1.2)$ — small overlap.
Larger KL divergence $\Rightarrow$ faster convergence of consumption shares.

Close vs. Far: Consumption Share Dynamics



Top row (close beliefs): convergence is *slow* — median paths linger near $\lambda = 0.5$.
Bottom row (far beliefs): convergence is *fast* — dashed vertical lines show the period at which the median share reaches near 0 or 1.
Within far beliefs, $D_{\mathrm{KL}}(f, g) > D_{\mathrm{KL}}(g, f)$ explains why convergence is faster when nature chooses f (left) than g (middle).

KL Divergence Summary: Close vs. Far

Case	$D_{\mathrm{KL}}(f,g)$	$D_{\mathrm{KL}}(g,f)$	$D_{\mathrm{KL}}(h,f)$	$D_{\mathrm{KL}}(h,g)$
Close	0.003	0.003	0.073	0.061
Far	0.759	0.344	0.073	0.281

Observations:

- Close case: $D_{\mathrm{KL}}(f,g) \approx D_{\mathrm{KL}}(g,f)$, slow and symmetric convergence.
- Far case: $D_{\mathrm{KL}}(f,g) \gg D_{\mathrm{KL}}(g,f)$; faster convergence when nature$=f$ than when nature$=g$.
- $D_{\mathrm{KL}}(h,g) > D_{\mathrm{KL}}(h,f)$ in both cases $\Rightarrow$ agent 1 is always closer to nature's mixture h.

Summary

1. **Likelihood ratio process** $l_t(s^t) = \pi_t^1(s^t)/\pi_t^2(s^t)$ is the key state variable linking beliefs to allocations.
2. **Social planner's rule:** agent 1's consumption share equals the continuation Pareto weight $\lambda_t = \lambda\, l_t/(1 - \lambda + \lambda\, l_t)$.
3. **Competitive equilibrium prices** are a Pareto-weighted average of agents' pricing kernels, converging to the smarter agent's kernel.
4. **Blume–Easley answer:** competitive markets do transfer wealth (and hence pricing power) to the better-informed agent, vindicating Alchian and Friedman — but the *rate* depends on KL divergences.
5. **KL divergence governs convergence speed:** the larger $D_{\mathrm{KL}}(h\|g)$ relative to $D_{\mathrm{KL}}(h\|f)$, the faster agent 1 accumulates consumption.

Related Lectures and References

Related QuantEcon lectures:

- Lecture 22: *Likelihood Ratio Processes* (foundations)
- Lecture 29: *Likelihood Ratio Processes and Bayesian Learning*
- Lecture 88: *Competitive Equilibria with Arrow Securities* (homogeneous beliefs)
- Lecture 89: *Heterogeneous Beliefs and Bubbles* (Harrison–Kreps)

Key references:

- Blume, L. and D. Easley (2006). "If you're so smart, why aren't you rich? Belief selection in complete and incomplete markets." *Econometrica* 74(4), 929–966.
- Alchian, A. (1950). "Uncertainty, evolution, and economic theory." *Journal of Political Economy* 58(3), 211–221.
- Friedman, M. (1953). *Essays in Positive Economics*. University of Chicago Press.

Exercises (Selected)

Exercise 1. Starting from the first-order condition $p_t(s^t) = \delta^t \pi_t^i(s^t)/(\mu^i c_t^i(s^t))$, derive

$$p_t(s^t) = \delta^t \lambda(1-\lambda)\big[(1-\lambda)\pi_t^2(s^t) + \lambda \pi_t^1(s^t)\big].$$

Exercise 2. Bayesian agents: each agent updates a prior over f vs. g.

- Set $f \sim \text{Beta}(1.5, 1)$ and $g \sim \text{Beta}(1, 1.5)$.
- Study consumption share dynamics and learning speed for different initial priors $\pi_0^1 \neq \pi_0^2$.

Exercise 3. Same as Exercise 2 but with more separated distributions: $f \sim \text{Beta}(2, 5)$, $g \sim \text{Beta}(5, 2)$. Compare convergence speeds with the close-beliefs case.

Exercises 4–5. Two agents, three possible models (f, g, h): study how extreme priors affect posterior evolution and consumption allocations.